Self-Interacting Dynamics with Exponential Memory: Markovian Embedding, Contractive Memory Fibres, and Metastability

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Abstract

We formulate a minimal class of self-interacting stochastic processes in which a visible state x_n is coupled to an explicit memory field ρ_n with exponential forgetting. The visible trajectory is generally non-Markovian: the next-step law depends on the retained memory field, not on x_n alone. The augmented state $z_n = (x_n, \rho_n)$, however, is a time-homogeneous Markov process. We derive the exact exponential memory expansion, prove pathwise contraction of the affine memory fibre along a common visible trajectory, and represent the dynamics by a Markov kernel and its positive unital observable operator. We also describe the associated random skew-product structure and a formal continuum approximation. Metastable "knots" are treated operationally as long-lived regions or slow modes diagnosed by residence times, retained memory weight, autocorrelation, local Ornstein–Uhlenbeck linearization, and finite-rank transfer-operator estimates. Claims about spacetime emergence, Lorentz kinematics, quantum dynamics, Standard-Model structure, physical masses, or fundamental constants are outside the scope of this anchor paper and are listed only as future-work directions.

I. INTRODUCTION

Self-interacting stochastic processes provide a natural mathematical language for systems whose future evolution is shaped by their own past. Reinforced random walks, self-interacting diffusions, true self-avoiding walks, and self-repelling Brownian polymers all instantiate the general pattern

$$\begin{aligned} &\text{trajectory} \longrightarrow \text{occupation history} \\ &\longrightarrow \text{self-induced drift.} \end{aligned}$$

Classical examples include self-interacting diffusions [1], reinforced random processes [2], Brownian polymers [3], true self-repelling motion [4], and true self-avoiding walks [5, 6].

The present paper isolates a particularly simple variant: the retained history is not the full local time or a Cesaro-averaged empirical measure, but an explicit exponentially relaxing field. This produces a finite effective memory scale and a direct Markovian embedding. The construction is also close in spirit to Markovian embeddings of memory equations, such as auxiliary variable treatments of generalized Langevin dynamics [7–9]; the difference is that the memory variable here is a self-induced spatial field generating the drift.

The purpose of this paper is deliberately narrow. We do not attempt to derive spacetime, relativistic kinematics, quantum mechanics, particle spectra, or physical masses. Instead, we establish the model's mathematical anchor:

$$\begin{aligned} &\text{define the model} \\ \implies &\text{ prove the Markov embedding} \\ \implies &\text{ make metastability measurable.} \end{aligned}$$

The result is a conservative framework in which later emergence claims can be posed as separate conjectures or companion papers rather than as assumptions inside the anchor model.

For repository-level orientation, the downstream tracks are explicitly separated: the propagation and spacetime-kinematics programme is maintained as Paper II at `https://github.com/MemoryDynamics/Knoten/tree/main/paper/paper_ii`, while the quantum and Standard-Model programme is kept as a planned Paper III roadmap at `https://github.com/MemoryDynamics/Knoten/tree/main/paper/paper_iii`. These links are pointers to companion workstreams only; no statement in this paper depends on them as evidence.

II. DISCRETE MODEL

Let $\Omega \subseteq \mathbb{R}^d$ be an effective state space with reference measure dx. The Euclidean structure is used only to write kernels, gradients, and diagnostics; it is not identified with physical space. Let G_σ be a smooth normalized deposition kernel of width σ, and let K be an interaction kernel for which the convolution

$$(K * \rho)(x) = \int_\Omega K(x, y)\rho(y)\, dy$$

is differentiable in the regimes considered below.

Assumption 1 (Baseline regularity)**.** *For the structural statements in Sections III–IV, it is enough that G_σ is a normalized nonnegative density and that the update equations are well-defined. We either take $\Omega = \mathbb{R}^d$ or impose boundary/extension conventions under which $x_{n+1} \in \Omega$ and the kernel deposition has the stated total mass. For continuum approximations one should assume enough smoothness and growth control on K and ∇K to make the drift locally Lipschitz or otherwise well posed in the chosen function* *space.*

The discrete process is

$$x_{n+1} = x_n + \varepsilon\xi_n - \eta\nabla\Phi_n(x_n), \tag{1}$$

$$\rho_{n+1}(x) = (1-\lambda_{\mathrm{m}})\rho_n(x) + \beta G_\sigma(x - x_{n+1}), \tag{2}$$

$$\Phi_n(x) = (K * \rho_n)(x), \tag{3}$$

where $0 < \lambda_{\mathrm{m}} < 1$, $\beta \geq 0$, $\varepsilon \geq 0$, $\eta \geq 0$, and ξ_n are independent centered noise variables, typically taken isotropic with unit covariance. The parameter λ_{m} is the forgetting rate, while β is the deposited memory strength. The previously used normalized convention is the special case $\lambda_{\mathrm{m}} = \beta = \alpha$.

Let $m_G = \int_\Omega G_\sigma(x)\,dx$ and $M_n = \int_\Omega \rho_n(x)\,dx$. Then

$$M_{n+1} = (1-\lambda_{\mathrm{m}})M_n + \beta m_G. \tag{4}$$

Thus the stationary total retained memory mass, when the scalar recursion is stable, is $M_* = \beta m_G/\lambda_{\mathrm{m}}$. Unit mass is preserved only in the normalized case $m_G = 1$, $M_0 = 1$, and $\beta = \lambda_{\mathrm{m}}$. If ρ_0, G_σ, and β are nonnegative, every ρ_n is nonnegative. The choice $\beta = \lambda_{\mathrm{m}}$ is therefore a useful density normalization, not a structural requirement; an overall memory scale can often be absorbed into K or η.

The project terminology calls long-lived localized feedback regimes "knots." In this paper the word has only an operational meaning: a knot is a metastable region or slow mode of the augmented dynamics, not a fundamental particle and not an exact fixed point.

III. EXPONENTIAL MEMORY AND CONTRACTIVE FIBRES

The memory equation can be unrolled exactly.

Proposition 1 (Exponential memory representation)**.** *For $n \geq 1$,*

$$\rho_n = (1-\lambda_{\mathrm{m}})^n \rho_0 + \beta \sum_{j=0}^{n-1} (1-\lambda_{\mathrm{m}})^j G_\sigma(\,\cdot - x_{n-j}). \quad (5)$$

Proof. The claim follows by induction. For $n = 1$ it is exactly Eq. (2). If it holds at n, then

$$\begin{aligned}\rho_{n+1} = {} & (1-\lambda_{\mathrm{m}})^{n+1}\rho_0 \\ & + \beta \sum_{j=1}^{n} (1-\lambda_{\mathrm{m}})^j G_\sigma(\cdot - x_{n+1-j}) \\ & + \beta G_\sigma(\cdot - x_{n+1}),\end{aligned}$$

which is Eq. (5) with $n+1$. $\square$

The deposited weights are

$$w_j = \beta(1-\lambda_{\mathrm{m}})^j. \tag{6}$$

Their relative tail is controlled by λ_{m}, independently of the deposition scale β. For a rest-weight threshold δ, the effective memory depth satisfies

$$j_\delta = \frac{\log\delta}{\log(1-\lambda_{\mathrm{m}})} \sim \frac{|\log\delta|}{\lambda_{\mathrm{m}}} \quad (\lambda_{\mathrm{m}} \ll 1). \tag{7}$$

Thus $\lambda_{\mathrm{m}}^{-1}$ is the natural update-count scale for memory persistence. Figure 1 shows the normalized convention $\beta = \lambda_{\mathrm{m}} = \alpha$ for representative values of α.

Proposition 2 (Pathwise memory-fibre contraction)**.** *Fix a common visible trajectory $x_1, \ldots, x_n$. Let ρ_k and $\tilde{\rho}_k$ solve Eq. (2) with the same deposited kernels but different initial memory fields. Then*

$$\rho_n - \tilde{\rho}_n = (1-\lambda_{\mathrm{m}})^n(\rho_0 - \tilde{\rho}_0). \tag{8}$$

Consequently, for any homogeneous norm on the

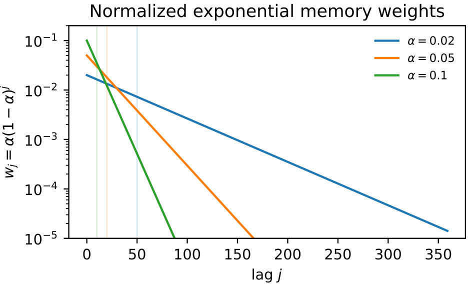

FIG. 1. Normalized exponential memory weights $w_j = \alpha(1-\alpha)^j$, corresponding to $\beta = \lambda_{\mathrm{m}} = \alpha$. Vertical guide lines mark $j = \lambda_{\mathrm{m}}^{-1} = \alpha^{-1}$ for each curve.

memory space,

$$\|\rho_n - \tilde{\rho}_n\| = (1-\lambda_{\mathrm{m}})^n \|\rho_0 - \tilde{\rho}_0\|. \tag{9}$$

Proof. Subtract the two memory recursions. The deposited kernels cancel because the visible trajectory is common:

$$\rho_{k+1} - \tilde{\rho}_{k+1} = (1-\lambda_{\mathrm{m}})(\rho_k - \tilde{\rho}_k).$$

Iterating gives Eq. (8). □

This contraction is a statement about the memory fibre under a fixed visible path. It is not yet a global contraction theorem for the full feedback system, because different memory fields can alter future positions through Eq. (1). A full contraction analysis must control the coupled pair (x_n, ρ_n).

IV. MARKOVIAN EMBEDDING

Define the augmented state

$$z_n = (x_n, \rho_n) \in \Sigma. \tag{10}$$

Given z_n and a fresh noise draw ξ_n, Eqs. (1) and (2) determine z_{n+1}. Thus the augmented dynamics is a time-homogeneous Markov process.

Definition 1 (Transition kernel). *For measurable $A \subseteq \Sigma$ define*

$$P(z, A) = \mathbb{P}\{z_{n+1} \in A \mid z_n = z\}. \tag{11}$$

The observable Markov operator is

$$(Uf)(z) = \int_\Sigma f(z')P(z, dz') = \mathbb{E}[f(z_{n+1}) \mid z_n = z]. \tag{12}$$

The operator U is linear, positive, and unital: $U1 = 1$. Its iterates satisfy $U^{m+n} = U^m U^n$. In the stochastic case it is generally not multiplicative: $U(fg)$ need not equal $(Uf)(Ug)$. The correct algebraic object is therefore a positive unital Markov operator on observables, with a dual action on probability measures, rather than a deterministic algebra automorphism.

Proposition 3 (Visible non-Markovian projection)**.** *The projected process x_n is generally not Markov.*

Proof. Take the same visible position x with two memory fields ρ and $\tilde{\rho}$ such that

$$\nabla(K * \rho)(x) \neq \nabla(K * \tilde{\rho})(x).$$

Then the conditional next-step laws of x_{n+1} differ because their drift terms in Eq. (1) differ. Hence conditioning on $x_n = x$ alone does not determine the next-step law. $\square$

The memory update should not be described as algebraically noninvertible in isolation. If x_{n+1} and

ρ_{n+1} are known and $\lambda_{\mathrm{m}} \neq 1$, then

$$\rho_n = \frac{\rho_{n+1} - \beta G_\sigma(\cdot - x_{n+1})}{1 - \lambda_{\mathrm{m}}}$$

is a formal rearrangement. The precise irreversibility statement is different: the present augmented state does not generally encode the full ordered history, and the stochastic dynamics defines a forward Markov semigroup rather than a reversible group action.

V. RANDOM SKEW-PRODUCT STRUCTURE

For each noise realization, write the update as

$$x_{n+1} = F_{\rho_n}(x_n, \xi_n), \tag{13}$$
$$\rho_{n+1} = T_{x_{n+1}}(\rho_n), \tag{14}$$

where

$$T_x(\rho) = (1 - \lambda_{\mathrm{m}})\rho + \beta G_\sigma(\cdot - x).$$

Equivalently, define the one-step augmented map

$$\mathcal{F}_\xi(x,\rho) = \big(F_\rho(x,\xi),\, T_{F_\rho(x,\xi)}(\rho)\big)\,.$$

Thus the system is a self-coupled random affine skew product with an exponentially contractive memory fibre along fixed visible paths. For a noise sequence $\omega = (\xi_0, \xi_1, \ldots)$ one obtains the random cocycle

$$\varphi(n,\omega,z_0) = \mathcal{F}_{\xi_{n-1}(\omega)} \circ \cdots \circ \mathcal{F}_{\xi_0(\omega)}(z_0),$$

in the standard sense of random dynamical systems [10]. This viewpoint is useful for discussing random attractors, path dependence, bifurcations, and metastability under noise, but the present paper uses it only as structural orientation.

VI. FORMAL CONTINUUM APPROXIMATION

Let $t = \lambda_{\mathrm{m}} n$ and consider a joint scaling

$$\lambda_{\mathrm{m}} \to 0, \qquad \varepsilon \to 0, \qquad \frac{\varepsilon^2}{2\lambda_{\mathrm{m}}} \to D,$$
$$\frac{\eta}{\lambda_{\mathrm{m}}} \to \gamma, \qquad \frac{\beta}{\lambda_{\mathrm{m}}} \to \kappa.$$

The final scaling assumption keeps the memory source finite; the normalized case has $\kappa = 1$. Formally, the visible coordinate approaches

$$dX_t = -\gamma \nabla (K * \rho_t)(X_t)\, dt + \sqrt{2D}\, dW_t, \tag{15}$$

while the memory field satisfies

$$\partial_t \rho_t = \kappa G_\sigma(\cdot - X_t) - \rho_t. \tag{16}$$

More generally one may introduce a memory time τ and write

$$\partial_t \rho_t = \tau^{-1} \left(\kappa G_\sigma(\cdot - X_t) - \rho_t \right),$$

equivalently

$$\rho_t = \kappa \tau^{-1} \int_{-\infty}^{t} e^{-(t-s)/\tau} G_\sigma(\cdot - X_s)\, ds$$

after transients. The extended process (X_t, ρ_t) is local in the scaled time variable and Markovian; the visible process X_t alone retains memory.

Equations (15)–(16) are a formal effective description, not a convergence theorem in this paper. A rigorous limit would require specifying topology, tightness estimates, regularity of K, boundary conditions, and the scaling of Ω if the domain changes with λ_{m}.

VII. METASTABILITY AND KNOT DIAGNOSTICS

A candidate knot is a long-lived localized feedback regime in the augmented dynamics. Operationally, for a region $\mathcal{K} \subseteq \Omega$ one can track retained memory mass

$$M_n(\mathcal{K}) = \int_{\mathcal{K}} \rho_n(x)\, dx. \tag{17}$$

Persistence over many memory times,

$$\Delta n \gg \lambda_{\mathrm{m}}^{-1},$$

is a minimal diagnostic for metastability, not a proof of a new particle-like object.

Several diagnostics are useful and mutually constraining:

- residence-time statistics of x_n in candidate regions;
- retained memory weight $M_n(\mathcal{K})$;

- autocorrelation functions of positions and reduced memory features;
- local linearization near a metastable center x_*;
- slow modes of a transfer-operator estimate on augmented features.

For local linearization, suppose a metastable regime has slowly varying memory ρ and a center x_* with $\nabla(K * \rho)(x_*) = 0$. With

$$H = \nabla\nabla(K * \rho)(x_*)$$

the continuum approximation becomes locally

$$dX_t = -\gamma H(X_t - x_*)\, dt + \sqrt{2D}\, dW_t. \tag{18}$$

Positive eigenvalues of γH give local relaxation rates in stable directions. A relaxation-based stability scale can be written as

$$\Gamma_{\mathrm{rel}} = \tau_{\mathrm{rel}}^{-1}, \qquad \widehat{\Gamma} = \frac{\sigma^2}{D}\Gamma_{\mathrm{rel}}.$$

This is a mass-like or stability proxy only. It is not a calibrated physical mass and should not be identified with mc^2.

A. Transfer-Operator View

Let $\psi(z_n)$ be a reduced feature vector for the augmented state, for example position, memory centroid, current offset from the memory centroid, memory spread, retained memory mass, or learned/PCA/Fourier features of ρ_n. A finite-rank Ulam or Markov-state approximation estimates a transition matrix at sample lag ℓ from pairs

$$(\psi(z_n), \psi(z_{n+\ell})).$$

Let τ_ℓ denote the update-time or scaled-time lag represented by those pairs. For finite trajectories, states that are observed only as terminal lagged targets have no empirical outgoing row. The reference implementation reports how many such rows are handled and, by default, treats them as absorbing in the finite matrix. Eigenvalues with modulus numerically equal to one are still reported in the spectrum but are not converted into relaxation rates. Production analyses should check the terminal-row count, vary the lag and discretization, and compare against independent residence diagnostics. Slow non-unit eigenvalues $\lambda_i(\tau_\ell)$ then give implied rates

$$\Gamma_i(\tau_\ell) = -\frac{1}{\tau_\ell} \log |\lambda_i(\tau_\ell)|. \tag{19}$$

Almost-invariant sets and coherent sets are then candidate metastable knots [11]. Related variational and Koopman approaches are standard in molecular kinetics and operator-theoretic model reduction [12–15].

Figure 2 shows one small reproducible transfer spectrum from the companion pipeline. It is an illustration of the diagnostic workflow, not evidence for a universal phase diagram.

VIII. REPRODUCIBLE NUMERICAL PIPELINE

The repository includes a minimal Paper-0 pipeline with the following steps:

1. run the finite-memory simulation over grids of noise, coupling, memory, kernel scales, kernel amplitudes, and dimension;

2. record reduced augmented-state features

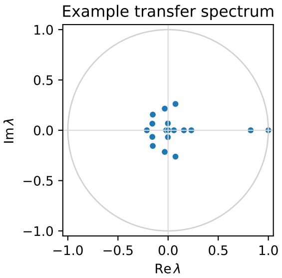

FIG. 2. Example eigenvalues of a finite-state transfer estimate from reduced augmented features. Non-unit modes close to the unit circle are candidates for metastable structure, subject to lag-time choice, discretization, finite-sample row handling, and sampling checks.

$\psi(z_n) = (x_n, \mathrm{summary}(\rho_n))$ at fixed sampling intervals;

3. compute residence statistics, covariance/occupancy diagnostics, and autocorrelation functions;

4. estimate a finite transition matrix on augmented features at sample lag ℓ;

5. report leading eigenvalues and implied rates $-\tau_\ell^{-1} \log |\lambda_i(\tau_\ell)|$;
6. validate candidate metastable sets against residence statistics and autocorrelation decay.

The reduced representation of ρ_n is intentionally labelled as a projection. It is a numerical approximation to the augmented state, not an exact replacement for the field. The point is to make metastability testable without storing a full field at every update.

IX. RELATION TO EXISTING LITERATURE

The closest mathematical family is self-interacting diffusion, where the drift depends on occupation-type memory variables [1]. The main difference is the explicit exponential forgetting in Eq. (2). Reinforced random walks [2], true self-avoiding walks [5], self-repelling motion [4], and Brownian polymers [3] share the feedback pattern but often emphasize repulsion or full/local-time memory.

Generalized Langevin theory and Mori–Zwanzig style embeddings [7, 8] provide the broader idea that nonlocal memory can be made local by adding variables. In the present model the added variable is not an auxiliary friction coordinate but the self-induced memory field itself.

Metastability is naturally formulated through transfer operators, Markov-state models, and Koopman approximations [12–15]. Large-deviation and noise-induced transition theory [16, 17] provides a separate asymptotic language for rare transitions once a small-noise regime has been specified.

X. OPEN MATHEMATICAL PROBLEMS

The structural results above do not settle the main hard questions. The following are open tasks:

- well-posedness of the continuum approximation under explicit kernel and state-space assumptions;
- Feller properties and invariant measures for the augmented Markov process;
- Lyapunov/Harris drift conditions of the form $PV \leq \lambda V + b$ outside compact sets [18];

- conditions for uniqueness, geometric ergodicity, and spectral gaps;
- coupled contraction estimates for (x_n, ρ_n), not only the memory fibre under a fixed path;
- bifurcation and metastability regimes for attractive, repulsive, and two-scale attractive-repulsive kernels;
- convergence and bias analysis for transfer-operator estimates on reduced memory features.

XI. BOUNDARIES AND FUTURE WORK

The following statements are not claims of this paper:

- that dimension $d = 3$ is selected by the model;
- that exponential memory alone implies a hard finite signal speed;
- that Lorentz invariance follows from the present equations;

- that quantum mechanics, $\hbar$, c, spin, gauge symmetry, or Standard-Model structure emerges here;
- that metastable knots are real particles;
- that relaxation rates are calibrated physical masses.

These questions may be posed later as falsifiable companion programs. A hard finite propagation bound, for example, would require additional assumptions such as finite interaction range, local multi-step transmission, and no direct long-range coupling term. Dimension-selection claims require controlled parameter sweeps, normalized kernel scaling, multiple seeds, and negative controls.

The current Paper II companion draft at `https://github.com/MemoryDynamics/Knoten/tree/main/paper/paper_ii` should therefore be read as a separate propagation and spacetime-kinematics programme rather than as an implication of the present model. The Paper III roadmap at `https://github.com/MemoryDynamics/Knoten/tree/main/paper/paper_iii` is intentionally only a future-program placeholder for quantum and Standard-Model questions.

XII. CONCLUSION

The model defines a self-interacting stochastic process with exponential forgetting and a field-mediated self-potential. The visible trajectory is generally non-Markovian, but the explicit memory field yields a Markovian augmented dynamics. The memory fibre is affine and pathwise contractive for a common visible trajectory, while feedback through the self-induced potential can support nontrivial metastable regimes. This is the mathematical anchor needed before stronger emergence claims can be formulated and tested.

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